

Anemia in children

Anemia is defined as reduction in red blood cell (RBC) mass, which will be measured in the laboratory, by reduction in hemoglobin concentration or hematocrit or RBC count.

WHO criteria for anemia. For children
6mo-59months <11g/dl, 5yrs-12yrs <11.5g/dl, 12-15yrs
<12g/dl, after 15 yrs. adult cut off points are used

Clinical features

- Common clinical manifestations include easy fatigability, shortness of breath, palpitation, dizziness, vertigo, irritability and loss of appetite

Physical examination

- may reveal pallor, tachycardia, atrophy of tongue papillae, swollen edematous tongue, systolic ejection murmur, hepatosplenomegaly, bone tenderness and
- extremity edema

Diagnosis

- guided by history, physical examination findings and the likely differential diagnosis considered accordingly

Investigation

Complete blood count,

- . Reticulocyte count
- . Stool examination.... For intestinal parasites, occult blood
- . Peripheral morphology

Types

- o Iron deficiency anemia.
- o Megaloblastic (pernicious) anemia.
- o Hemolytic anemia.
- o Sickle cell anemia.
- o Cooley's anemia (thalassemia).
- o Aplastic anemia
- o Chronic anemia.
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Evaluation of the anemic child

- first step Make sure a child is truly anemic
- The second step
- Developing a differential diagnosis
- History, physical examination, CBC and the peripheral blood smear (PBS)

- Step 1 : Determine the child is anemic using the Hb/Hct
- Step 2: Evaluate the Red Cell Indices
 - Mean corpuscular volume (MCV)
 - The lower limit for the MCV(<10 years) approximately $70 \text{ fL} + \text{age in years}$
 - The upper limit for (>0.5 years) is $84 + 0.6 \text{ fL per year}$ until the upper limit of 96 fL
 - Enables classification of anemia
 - microcytic
 - normocytic or
 - macrocytic.

- MCV = is the measure of the volume of the "average" red blood cell, stated in femtoliters (fL).

Microcytic anemia

iron deficiency

- lead poisoning,
- anemia of inflammation, or
thalassemia (or other more rare hemoglobinopathies)

- Iron deficiency anemia
- is the most common
- age of the patient and diet history raise the index of suspicion for the diagnosis
- Peak prevalence occurs during late infancy and early childhood
- A second peak of iron deficiency is seen during adolescence In these populations
 - is due to (in both cases) high growth rate
- a therapeutic trial of oral iron is an appropriate initial diagnostic test

Risk of iron deficiency

- Perinatal factor: prematurity, low birth weight, fetomaternal hemorrhage, TTTS
 - Dietary factors: low iron bioavailability of iron (cow's milk), poor iron content of diets, diets which decrease bioavailability of iron
 - . Chronic blood loss:
 - . Decreased absorption: celiac disease, surgery, CD
 - . Adolescent girls also are at higher risk of iron deficiency because their bodies lose iron during menstruation.

Pharmacologic treatment of iron deficiency anemia

- Oral iron replacement

First line

- Ferrous sulphate, elemental iron 3-6mg/kg/day, treatment has to be continued for 2-3months after Hemoglobin has normalized

Macrocytic anemia

- Due to a relative decrease in DNA synthesis during erythropoiesis
 - Folate and/or vitamin B12 deficiency
 - Ineffective erythropoiesis or marrow failure
 - Liver disease and hypothyroidism

Pharmacologic treatment of megaloblastic anemia

- Folate deficiency Folic acid, 1 to 5 mg p.o. daily for 1- 4 months,

Vitamin B12 Deficiencies

- nutritional deficiency of vitamin B12 is extremely rare except in strict vegans (strict vegetarian)
deficiency is due to malabsorption
Terminal Ileum resection
Pernicious anemia (intrinsic factor deficiency)
inherited disorders of transport or metabolism of vitamin B12

Pharmacologic treatment of megaloblastic anemia –

- Cyanocobalamin (Vitamin B12), 1000 micrograms (1mg), IM, every day for one week, every week for four weeks and then, if the underlying disorder persists, 1 mg every month for the remainder of the patient's life.

Folate Deficiency

- Dietary
- The common dietary sources of folate are green leafy vegetables
- Breast milk and infant formulas provide adequate folate
- Folate deficiency can be found in infants and children who primarily consume **goat's milk**